

10

Value at Risk

10.1 Introduction

One of the key risk management concepts in use in financial markets today is Value-at-Risk (VaR). Value-at-Risk is a measure of the maximum potential change in value of a portfolio of financial instruments with a given probability over a pre-set horizon. VaR answers the fundamental question facing all risk managers: How much can I lose with a given probability over a chosen number of days? The concept of VaR is a very appealing one – it can be developed for any kind of portfolio and can be aggregated across portfolios containing different kinds of instruments. One of the major advantages of VaR is that it has become a widely accepted industry standard with support from such bodies as the Group of Thirty, The Bank of International Settlements, and the European Union.

Our objective in this chapter is to give the reader sufficient understanding of the implementation of the main VaR approaches as they relate to energy markets to be able to build a basic VaR measurement system. We begin our discussion of VaR, in the following section, by introducing the basic concepts of VaR. In the following two sections we look at the basic model of energy market variables returns used in VaR and methods of forecasting the key parameters from historical data – the volatilities and correlations of energy spot and forward prices. We then go on to describe in detail the implementation of the four main VaR approaches – delta, delta-gamma, historical simulation and Monte Carlo simulation. For each of these methods we give a numerical example of calculating the VaR for an energy derivatives portfolio containing crude oil and natural gas futures options. The testing of the accuracy of the VaR forecasts is the subject of the penultimate section and the final section contains concluding remarks.

Throughout this chapter we reference the RiskMetrics¹ Technical Document (J.P. Morgan (1996)) which is an excellent source of further information on Value-at-Risk methods. We also recommend Duffie and Pan (1997) and Shimko (1997a, 1997b).

10.2 What is Value at Risk?

VaR is a measure of the level of loss which would not be expected to be exceeded over a chosen time period (the VaR horizon which we denote T_h) with a chosen confidence level (or probability which we denote p) under normal market conditions. Let us consider the

¹ RiskMetrics is a registered trademark of J. P. Morgan.

key components of this statement carefully. VaR is stated as a level of loss for a chosen confidence level and horizon – e.g. USD 2 million over ten days at 95%. The VaR is quoted in terms of percentage loss. This means that we would expect our loss on the portfolio, over a ten day period, to exceed USD 2 million only 5% of the time or once in every twenty independent ten day periods. Note that the VaR figure is not an absolute worst case or maximum loss and that the VaR is specific to a particular portfolio, the horizon and the confidence level chosen. In particular the VaR will be greater for higher confidence levels – the reasons for this will become clear later in the chapter. The calculation of the VaR implicitly assumes that the portfolio can be liquidated at the VaR level without additional costs, this implies normal market and trading conditions. If the market has just undergone a large correction then the costs, in terms of spreads, of liquidating any reasonably sized portfolio could be significant. Thus we can re-state the definition of VaR as the loss that is expected to be exceeded with probability $1-p$ in a T_h year period under normal market conditions.

Underlying VaR is the concept of a probability distribution of portfolio values at the VaR horizon. Figure 10.1 illustrates a probability distribution for a portfolio over a ten day horizon which has a current value of \$10 million and a VaR of \$2 million at a confidence of 95%.

The probability, or confidence, of getting a particular range of values for the portfolio is given by the area under the curve in Figure 10.1 – this is roughly the portfolio value

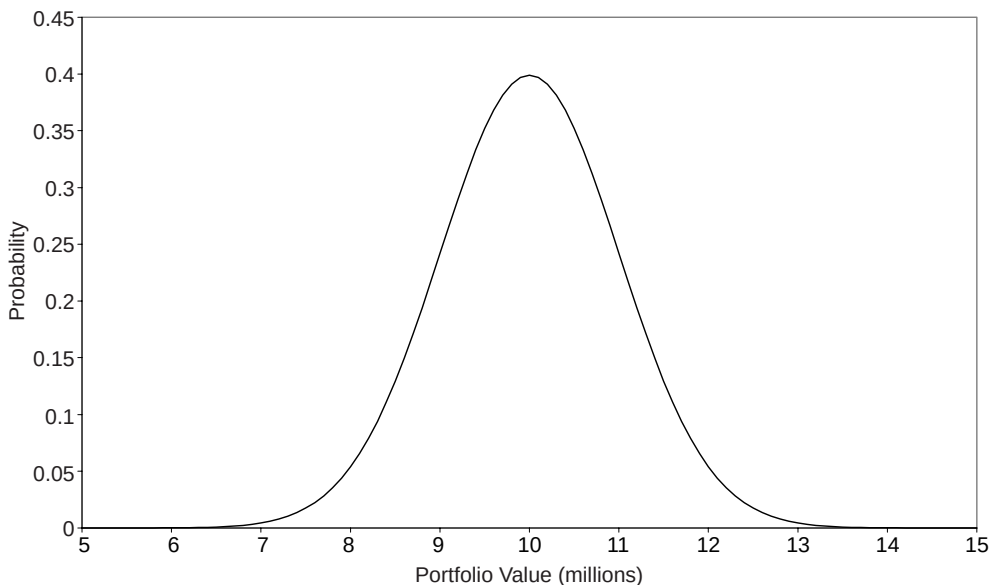


FIGURE 10.1 Illustration of a Typical Portfolio Distribution

range multiplied by the average probability over that range. Since we are interested in losses above a certain level, we are interested in the far left end of the curve which is called the left-hand tail. The area under the curve from its left-most point (zero) up to \$8 million (that is the \$10 million current value less a \$2 million loss) is approximately 5% (this is called a percentile) and therefore we are 95% confident we will not get a loss worse than \$2 million in the next ten days with this portfolio.

Because VaR looks at the risk of the entire portfolio, it is an integrated way to deal with different markets and different sources of risk. It allows all these different factors to be combined into a single number – one which is a good indicator of the overall risk level. However, it is not a sufficient risk measure for risk management and risk control. It is also important to emphasise that VaR is only an estimate of the potential loss.

One of the great strengths of VaR is that it provides a ‘global’ view of risk, and as such has the potential to be used as the basis of integrated risk management. An energy trading firm may simultaneously trade many energies such as oil, gas, and electricity as well as having cash and foreign exchange positions. Because of the substantial correlation between energy prices, the overall risk to the firm should be much less if all the energies are treated together in one book, rather than divided into separate books which are treated individually. VaR provides a way to net the price risks across different energy markets and so achieve more efficient controls – position limits, capital requirements, etc. New deals can be assessed for their impact on the firms total risk using the VaR approach (see Garman (1997) for efficient ways to achieve this type of analysis). Finally, a main use of the VaR value is as an estimate of the level of capital required to support the trading or other operations of the company. In this way it can be used to assess the efficiency with which the capital is being used (see Shimko (1997a)).

10.3 Modelling the Market Variables

The first step in evaluating the risk of a portfolio, as with pricing and hedging, is to define how the underlying market variables - energy prices, interest rates, exchange rates, etc. might change in the future.² This requires that we specify a model for these market variables. The model used in the basic VaR methods is that market variable returns are normally distributed, also called the random walk model - we discuss the shortcomings of this model in the sections on historical and Monte Carlo simulation and in our conclusions in section 10.9. The simple mean reverting spot price model of chapter 2, the models of chapter 6 except for the model which incorporates jumps, and the general multi-factor model in chapter 8 are consistent with the assumption of normally distributed underlying market variables³. As usual we define returns as the change in the natural logarithm of the market variable over a certain period of time, i.e. if $S_{t-\Delta t}$

² In this chapter we will focus on energy spot and forward prices as the underlying market variables.

³ Under the mean reverting spot price model, stochastic convenience yield and/or stochastic interest rates and multi-factor forward curves models, both spot prices and forward or futures prices are normally distributed. However, we must be careful that the volatility is calculated in an appropriate way – we indicate this at the appropriate points in this chapter (see also chapter 3).

and S_t are the market variable values at times $t - \Delta t$ and t then the return, which we denote by x_t , is given by;

$$x_t = \ln(S_t) - \ln(S_{t-\Delta t}) \quad (10.1)$$

The random walk model can be represented as follows;

$$x_t = \mu_t + \sigma_t \varepsilon_t; \varepsilon_t \sim \text{IID } N(0, 1) \quad (10.2)$$

This simple model has two parameters associated with each individual market variable - the mean μ_t and variance σ_t^2 of the normal distribution. The mean is usually assumed to be zero as it does not generally have a large effect on VaR estimates⁴. The square root of the variance is the standard deviation which is often referred to as the volatility. It has been claimed in numerous articles that this model is inappropriate for commodities and energies because it does not allow for mean reversion. This is only true if the model is interpreted in a naive way where the underlying market variables are forward prices with fixed maturity dates. We saw in chapter 8 that forward prices with fixed maturity dates have increasing instantaneous volatility as they approach maturity and that this is directly related to the mean reversion in the spot price. However, forward prices of constant time to maturity will generally have approximately constant volatility⁵. Furthermore, we saw in chapter 8 that the volatility of a constant maturity date forward price over a specified time period, in the sense of the standard deviation of the return distribution, is given by the time average of the instantaneous forward price volatility over that time period. Thus equation (10.2) can even be applied to fixed maturity date forward price returns if we compute σ_t as the average volatility over the range of the maturities which the forward will have in the time period Δt . We highlight these points again in the relevant sections of this chapter.

The model represented by equation (10.2) also has a parameter associated with each pair of market variables, the correlation, which measures how closely the two variables move together. A correlation of 1 indicates the variables move exactly together, a correlation of zero indicates there is no association between their movements and a correlation of -1 indicates the two variables move in exactly opposite directions. In the following sections we discuss how these parameters can be estimated from historical market data.

10.4 Forecasting Standard Deviations (Volatility)

Since the VaR estimate is a forward looking estimate we need to forecast the values the parameters of our market model should take for the chosen VaR horizon. The simplest and most widely used method of forecasting the standard deviation of returns is the Simple Moving Average (SMA) method (also known as the rolling (sample) standard

⁴ See RiskMetrics Technical Document, Chapter 5, Section 5.3, page 90.

⁵ Neglecting the time varying component which we associate with the time varying spot volatility.

deviation). Imagine we have a history (or time-series) of returns of a market variable, for example a spot price or a constant maturity forward or futures price⁶, x_i , $i = 1, \dots, N$, with x_1 being the oldest return and x_N being the most recent return. The SMA estimate of the standard deviation $\hat{\sigma}$ (where $\hat{\cdot}$ signifies an estimate of the parameter σ) is given by;

$$\hat{\sigma} = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2} \quad (10.3)$$

This method of estimation has the disadvantage that all the returns, used to compute the estimate, have equal importance or weight. Thus a large return, which may be several months old, will affect the estimate just as much as a large recent return, even though the older return is associated with an event whose effect would have declined significantly. The Exponentially Weighted Moving Average⁷ (EWMA) estimator solves this problem by giving older returns exponentially less weight. The EWMA estimator is given by;

$$\hat{\sigma} = \sqrt{\frac{1}{\sum_{i=1}^N \lambda^{i-1}} \sum_{i=1}^N \lambda^{i-1} (x_i - \mu)^2} \quad (10.4)$$

The sum of the weighting factors can be simplified as follows;

$$\sum_{i=1}^N \lambda^{i-1} = \frac{1 - \lambda^N}{1 - \lambda} \cong \frac{1}{1 - \lambda} \quad (10.5)$$

Substituting the approximation⁸ in equation (10.5) into equation (10.4) gives the formula for the EWMA used in RiskMetrics;

$$\hat{\sigma} = \sqrt{(1 - \lambda) \sum_{i=1}^N \lambda^{i-1} (x_i - \mu)^2} \quad (10.6)$$

The EWMA estimator depends on the parameter λ ($0 < \lambda < 1$) which is called the decay factor and which determines the relative weight given to older returns. Figure 10.2 illustrates the weights which are applied to each sample return for a range of decay factors.

For decay factors close to 1, for example 0.99, the weighting on each return declines only slowly giving recent returns roughly the same weighting. For lower values of λ the

⁶ It is important that we use forward or futures prices which have a constant time to maturity because, as we saw in chapter 8, apart from seasonal effects, their volatility would be expected to be approximately constant.

⁷ This is the method used by RiskMetrics (see the RiskMetrics Technical Document, Chapter 5).

⁸ Note that this approximation is only good for typical values of λ . As λ approaches 1 the approximation fails e.g. $\lambda = 0.99$ gives weighting factors up to approximately 20% different from the correct values.

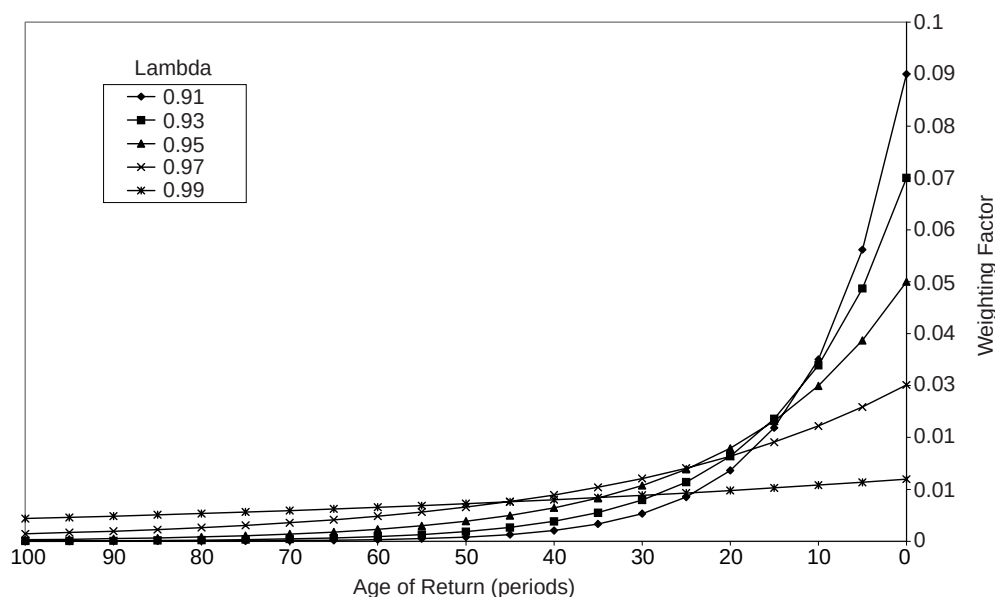


FIGURE 10.2 EWMA Weighting Factors for a Range of Typical Decay Factors (λ)

weightings decline much faster resulting in more recent returns contributing relatively more to the estimate. Note also that the number of returns which effectively contribute to the EWMA estimate depends on λ , for a value of 0.91 the weights beyond the 100th return are less than 10^{-5} , whereas for $\lambda = 0.95$ the weights only fall below 10^{-5} at the 170th return.

Figure 10.3 illustrates the difference between the SMA (using a 30 day sample size) and EWMA forecasts ($\lambda = 0.94$) on Brent Crude Oil spot prices for 1998.

Notice how both the SMA and EWMA forecasts jump on the 23 March 1998 due to the spot price going from \$13.04 to \$14.80 per barrel. The SMA forecasts then stays roughly constant for the following 30 days until the large return drops out of the 30 day moving window at which point the SMA forecast drops suddenly back to a lower level. In contrast the EWMA forecast slowly declines from the peak on the 23 March down to the lower level. The level of the decay factor used in Figure 10.3 is that recommended by RiskMetrics for daily time series.

Another attractive feature of the EWMA estimator is that it can be written recursively in terms of the previous days estimate if we assume that the sample mean of returns is zero. Thus the estimate on date $i + 1$ is given in terms of the estimate on date i by

$$\hat{\sigma}_{i+1} = \sqrt{\lambda \hat{\sigma}_i^2 + (1 - \lambda)x_i^2} \quad (10.7)$$

The decay factor λ determines how fast the effect on the estimate of large returns, such as that on the 23 March in Figure 10.3, declines. How should the decay factor be chosen?

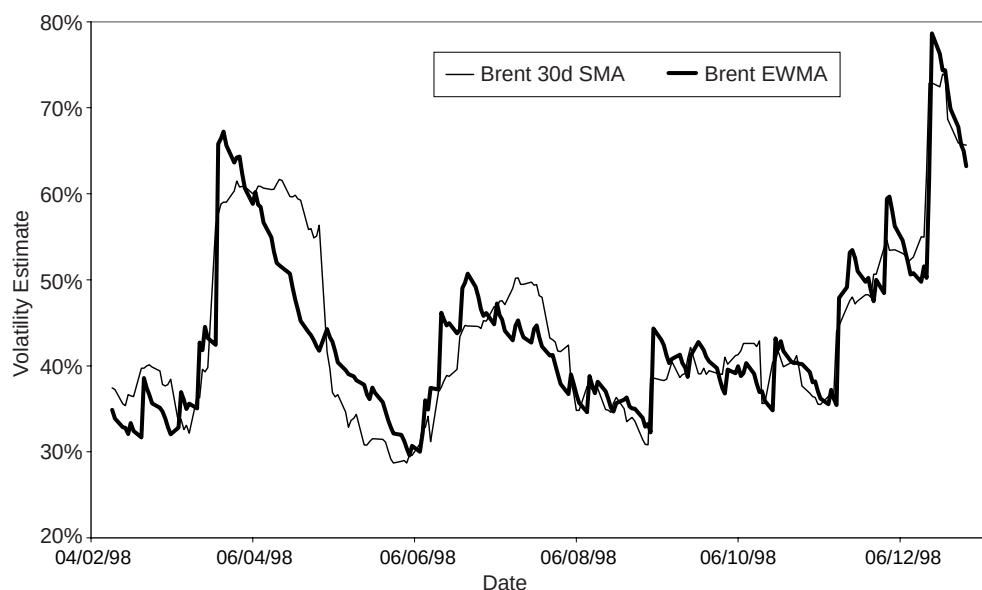


FIGURE 10.3 Comparison of a SMA and EWMA Standard Deviation Forecasts on Brent Crude Oil Spot Prices for 1998

Riskmetrics uses 0.94 for daily forecasts and 0.97 for monthly forecasts⁹. These optimal decay factors are obtained as the weighted average of the optimal decay factors for each market variable in the RiskMetrics data set. The optimal decay factors for each market variable are obtained by minimising the Root Mean Squared Error (RMSE) between the EWMA variance forecast for period $i + 1$ (given returns up to and including period i) and the actual squared return for period $i + 1$. Table 10.1 illustrates the results of computing the optimal decay factor for Brent crude oil and Henry Hub natural gas daily spot price returns for the periods from January 1995 to February 1999 and January 1998 to February 1999.

The optimal daily forecast decay factor for crude oil is much higher than the RiskMetrics value. Furthermore, the optimal daily forecast decay factor for natural gas is very different for the two periods. This is primarily due to the period of very high volatility in natural gas prices in the first quarter of 1996. This analysis indicates that it is important to independently obtain the optimal decay factors for energy data and not use the RiskMetrics standard values.

The time varying or seasonal behaviour of volatility is significant in many energy markets and it is therefore important to not only compute the optimal decay factor for each time series over a number of seasonal cycles but also to interpret the results. The RMSE indicates the average error in the EWMA variance forecast over the chosen

⁹ See section 5.3.2 of the RiskMetrics Technical Document, Fourth Edition.

TABLE 10.1 Optimal EWMA Decay Factor for Crude Oil and Natural Gas (Computed by Minimisation of the RMSE of the Variance Forecast)

Data Set	Brent Crude Oil		Henry Hub Natural Gas	
	RMSE	λ	RMSE	λ
Jan 95–Feb 99	0.001162	0.971131	0.056728	0.779144
Jan 98–Feb 99	0.001861	0.979190	0.010669	0.915702

period. If this error is a significant fraction of the actual average variance for the period this indicates that a single decay factor may not appropriate for the full seasonal cycle and one should look at estimating separate decay factors for the different “seasons”.

10.5 Forecasting Covariances and Correlations

The SMA and EWMA estimators can be used to construct covariance and correlation forecasts in a similar way to the forecasts of volatility. Imagine we have two time series of returns $x_{1,i}$ and $x_{2,i}$, $i = 1, \dots, N$, for two different market variables, for example crude oil and natural gas spot prices. The covariance forecast for the SMA and the EWMA are given by:

SMA:

$$\hat{\sigma}_{1,2}^2 = \frac{1}{N} \sum_{i=1}^N (x_{1,i} - \mu_1)(x_{2,i} - \mu_2) \quad (10.8)$$

EWMA:

$$\hat{\sigma}_{1,2}^2 = (1 - \lambda) \sum_{i=1}^N \lambda^{i-1} (x_{1,i} - \mu_1)(x_{2,i} - \mu_2) \quad (10.9)$$

Assuming μ_1 and μ_2 are zero, the EWMA covariance forecast can also be written in recursive form:

$$\hat{\sigma}_{1,2,i+1}^2 = \lambda \hat{\sigma}_{1,2,i}^2 + (1 - \lambda) x_{1,i} x_{2,i} \quad (10.10)$$

The standard definition of correlation is the ratio of the covariance of the two time series to the product of the standard deviations of the individual time series. The correlation forecast is therefore given by:

$$\hat{\rho}_{1,2,i} = \frac{\hat{\sigma}_{1,2,i}^2}{\hat{\sigma}_{1,i} \hat{\sigma}_{2,i}} \quad (10.11)$$

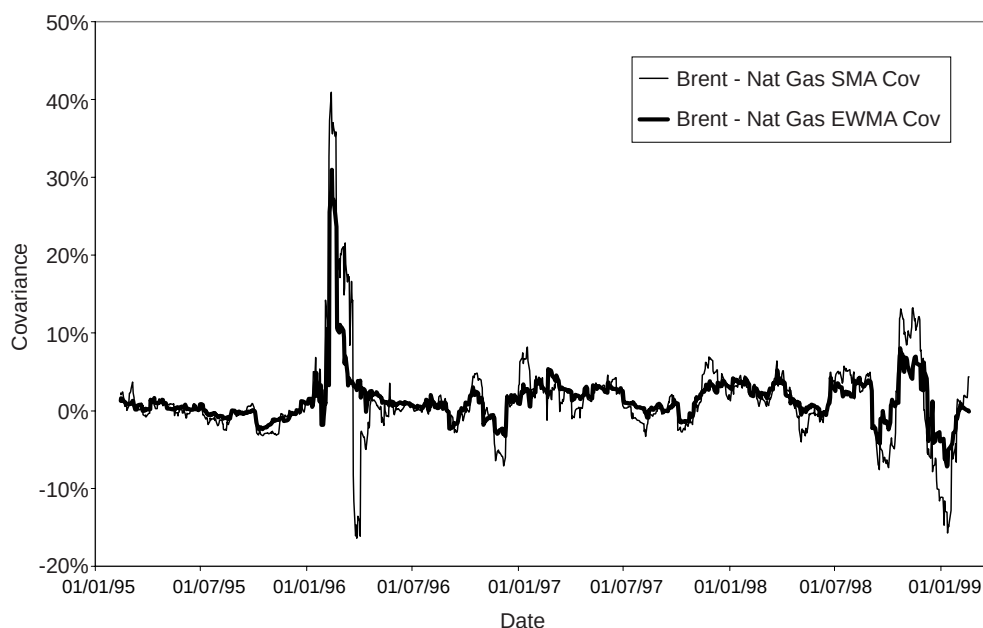


FIGURE 10.4 Comparison of a SMA and EWMA Covariance Estimators on Brent Crude Oil Spot Prices and Henry Hub Natural Gas Spot Prices for 1998

Figure 10.4 illustrates the SMA and EWMA estimators of the covariance between Brent crude oil spot prices and Henry Hub natural gas spot prices for 1998. Notice, in figure 10.4, how the EWMA forecast is more stable than the SMA forecast. Figure 10.5 shows the corresponding correlation forecasts.

10.6 Value at Risk Methodologies

In this section we describe the four main methods for computing VaR; Delta, Delta-Gamma, Historical Simulation and Monte Carlo Simulation. For each method we give a detailed numerical example using the same example portfolio containing an oil and natural gas futures option. Our objective here is to describe the key implementation steps and to highlight the pros and cons of these main approaches to VaR. The two main steps for calculating VaR are:

- All positions must be marked to market (valuation).
- Estimate the future variability of the market value (risk evaluation).

The following simple examples illustrate the basic approach to computing VaR and will aid in establishing the key concepts. The examples use standard deviations and correlations of market variable returns and the assumption that these returns are normally distributed.

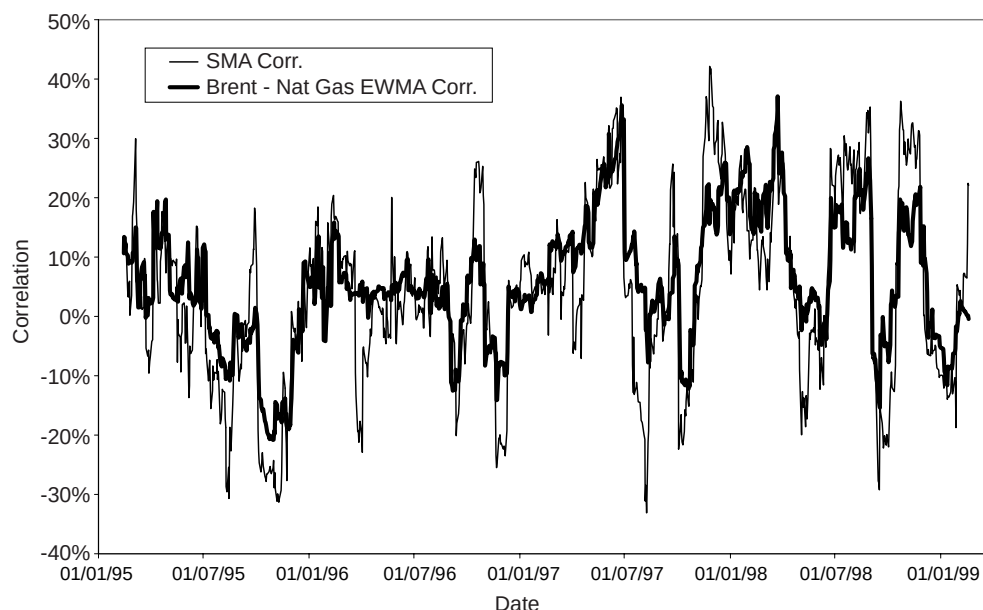


FIGURE 10.5 Comparison of a SMA and EWMA Correlation Estimators on Brent Crude Oil Spot Prices and Henry Hub Natural Gas Spot Prices for 1998

Example 1: A Single Underlying Market Variable – The Crude Oil Price

Consider a USD-based energy company with a USD 100 million spot crude oil position. They wish to calculate their VaR over a 1-day horizon such that there is a 5% chance that the realised loss will be greater than the VaR level (i.e. a confidence level of 95%). The 5% probability level may be a choice made by the energy company which reflects its appetite for risk or may be imposed by regulatory requirements. The first step in the calculation is to compute the exposure to market risk (i.e., mark-to-market the position). As a USD-based company, the exposure is equal to the market value of the position in the base currency i.e. USD 100 million.

The next step is to evaluate the risk of the position. In order to measure the risk we need an estimate of how much the crude oil price can potentially change over the selected horizon (in this case 1 day). The standard deviation of the daily returns of the crude oil price, measured historically, provides a measure of the size of the possible daily movements¹⁰. Imagine that the company calculates the daily standard deviation of the crude oil price to be 2.5%. Assuming that the returns on crude oil are normally distributed then, for the chosen confidence level of 95%, the VaR is given by –1.65

¹⁰ By computing the historical standard deviation over the same returns period as the VaR horizon we are implicitly taking into account the effect of mean reversion and also approximately jumps (see chapter 8).

times the standard deviation¹¹, that is $-1.65 \times 2.5\%$ or -4.125% . This implies that the crude oil price would not be expected to drop more than 4.125% in one day 95% of the time. In USD, the VaR of the position is equal to the market value of the position times the VaR percentage i.e. $\$100 \text{ million} \times -0.04125 = -\4.125 million . This means that 95% of the time, or 95 days out of 100, the energy company would not expect to lose more than \$4.125 million in that 24 hour period.

Example 2: Two Underlying Market Variables – The Crude Oil Price and the USD/BOL FX Rate

Consider a Venezuelan Bolivar (BOL) based energy company with a USD 100 million spot crude oil position. What is the VaR over a 1-day horizon such that there is a 5% chance that the realised loss will be greater than the VaR level (i.e. a confidence level of 95%)? The exposure is the same as in the previous example but the company now also has foreign exchange risk.

In order to evaluate the risk of the position we first calculate the VaR of the individual positions. The crude oil risk is the same as in the previous example but now we state it in the new base currency by converting at the current exchange rate of 500 BOL/USD giving $500 \times -4,125,000 = \text{BOL } -2062,500,000$. Imagine that the company calculates the daily standard deviation of the BOL/USD exchange rate to be 0.5%. Assuming that the returns on the exchange rate are normally distributed then the VaR for the foreign exchange position is $-1.65 \times 0.005 \times 500 \times 100,000,000$ or $\text{BOL } -412,500,000$. Now, the total risk of the crude oil position is not simply the sum of the spot oil price risk and the FX risk because of the correlation between the return on the oil price and the exchange rate. If the correlation is estimated to be -0.25 the total risk of the position is given by:

$$\begin{aligned} \text{Total VaR} &= \sqrt{\text{VaR}_{OIL}^2 + \text{VaR}_{FX}^2 + 2\rho_{OIL,FX}\text{VaR}_{OIL}\text{VaR}_{FX}} \\ \text{Total VaR} &= \\ &\sqrt{2062,500,000^2 + 412,500,000^2 + 2 * (-0.25) * 2062,500,000 * 412,500,000} \\ \text{Total VaR} &= \text{BOL } 1999,668,000 \end{aligned}$$

This is the simplest method of computing VaR and is called Variance-Covariance or Delta VaR. The application of this method to portfolios containing derivatives is described in detail in the next section.

In general, an energy portfolio will generate cash flows at a large number of different future dates. If we were to represent these cash flows explicitly we would potentially need a very large correlation matrix whose size would vary depending on the particular portfolio. Furthermore, there is generally a limited set of reliable correlation information (i.e. liquid forward/futures prices) in many energy markets. Therefore the Risk-

¹¹ We call this NSD – the number of standard deviation for a particular confidence level.

Metrics datasets provide volatility and correlation information for a predefined set of standardised maturities called the RiskMetrics Vertices – 1, 3, 6, 12 months, 2 to 5, 7, 9, 10, 15, 20 and 30 years. In RiskMetrics, cash flows not falling on these maturity dates are handled by mapping them onto the standard maturities. The mapping process involves splitting a cash flow into two equivalent cash-flows on the two standard maturities either side of the actual cash-flows maturity. Equivalence is defined by the two cash-flows having the same market value and risk (volatility) as the original cash flow (see Section 6.2, RiskMetrics Technical Document). An alternative is to use the volatility function or eigen vector representation of the volatility and correlation information. This allows us to obtain the volatilities and correlations for any pair of forward/futures contracts by interpolating from the volatility functions (see chapter 8 for details). The approach of only explicitly representing a limited number of maturities is important in practice for reducing the amount of data which must be analysed and stored in a VaR system.

In the following sections we describe in detail the implementation of Delta, Delta-Gamma, Historical Simulation, and Monte Carlo Simulation.

10.6.1 Variance-Covariance (Delta VaR)

The Variance-Covariance approach is that used in the two examples in the previous section. It assumes that the returns for all the market variables (e.g. energy prices, exchange rates, interest rates) are normally distributed. With this assumption the future possible values of the market variables can be captured by the Variance-Covariance Matrix (VCM) of the market variables returns. The VCM is computed from historical returns of forward/futures price data as described in section 10.4 (the RiskMetrics datasets gives this information for a wide range of key market variables although the energy market information is limited). Derivatives are handled by representing them in terms of a Delta equivalent position in the underlying asset (the Delta-Gamma VaR approach described in the next section is more accurate but more complex to implement).

The major advantage of this approach is that in many markets it is widely used and accepted as the basic method for evaluating VaR for portfolios without a large component of options. It is also the simplest method to understand and implement, and is the least computationally demanding. The disadvantages are that the underlying model of the market is not very good, particularly for energy markets (see section 10.9 for a discussion of the problems with the assumption of normally distributed returns) and it is not suitable for portfolios of options. Also, the volatilities and correlations are based on past history and may not reflect the future behaviour of the market (in fact this is true of all the main VaR approaches – we discuss a way to correct this problem in the concluding section 10.9). However, it is useful to study the details of Delta VaR before looking at the more advanced and complex methods. The first step is to represent the portfolio changes in terms of the changes in the underlying market variables which we are modelling.

Representing the Portfolio Returns in Terms of the Underlying Returns

Consider a portfolio with value P containing N instruments of value $V_i, i = 1, \dots, N$. The value of the portfolio is simply the sum of the individual instrument values;

$$P = V_1 + V_2 + \dots + V_N \quad (10.12)$$

The change in the value of the portfolio, ΔP , over the VaR horizon is given by

$$\Delta P = \Delta V_1 + \Delta V_2 + \dots + \Delta V_N \quad (10.13)$$

where ΔV_i are the changes in the value of the individual instruments over the VaR horizon. The return on the portfolio is therefore given by

$$\Delta P/P = (\Delta V_1 + \Delta V_2 + \dots + \Delta V_N)/(V_1 + V_2 + \dots + V_N) \quad (10.14)$$

If the instruments are all underlying market variables (spot or forward/futures prices) then $V_i = S_i$ and we can write the portfolio return in terms of the returns on the individual instruments as follows:

$$\Delta P/P = w_1 x_1 + w_2 x_2 + \dots + w_N x_N \quad (10.15)$$

Where the returns are given by the approximation¹² $x_i = \Delta S_i/S$ and $w_i = S_i/P$ are called leverage weights (see also Section 6.3.1.2, RiskMetrics Technical Document). Note that the portfolio return is not simply equal to the sum of the returns of the individual instruments (x_i). If the instruments are derivatives or non-linear instruments then we represent the change in the value of the instrument in terms of a delta ($\frac{\partial V}{\partial S}$) position in the underlying market variable:

$$\Delta V = \frac{\partial V}{\partial S} \Delta S \quad (10.16)$$

The return on the portfolio becomes

$$\Delta P/P = \frac{\partial V_1}{\partial S_1} w_1 x_1 + \frac{\partial V_2}{\partial S_2} w_2 x_2 + \dots + \frac{\partial V_N}{\partial S_N} w_N x_N \quad (10.17)$$

Note that equation (10.17) applies to the case of market instruments which are underlying market variables since in this case $\frac{\partial V}{\partial S} = 1$.

¹² The length of the return interval for which the approximation is good depends on the level of spot volatility, the rate of mean reversion and the volatility of long maturity forward prices. For typical energy parameter values, for example spot volatility of 35%, mean reversion rate of 10 and long maturity forward volatility of 20% the approximation is usually good for intervals up to a month and is certainly good for daily returns.

Computing Delta VaR

We now have the portfolio return represented in terms of a linear combination of underlying market variable returns. Since we are assuming that the returns are normally distributed we can write down the standard deviation of the portfolio returns in terms of the standard deviations of the underlying market variables returns. The VaR estimate is then simply the portfolio standard deviation multiplied by the number of standard deviations (NSD) appropriate to the chosen confidence level. The portfolio standard deviation is given by

$$\sigma_P = \Sigma R \Sigma^T \quad (10.18)$$

$$\Sigma = \left| \frac{\partial V_1}{\partial S_1} w_1 \sigma_1, \frac{\partial V_2}{\partial S_2} w_2 \sigma_2, \dots, \frac{\partial V_N}{\partial S_N} w_N \sigma_N \right| \quad (10.19)$$

$$R = \begin{vmatrix} 1 & \rho_{1,2} & \dots & \rho_{1,N} \\ \rho_{2,1} & 1 & \dots & \rho_{2,N} \\ \dots & \dots & \dots & \dots \\ \rho_{N,1} & \rho_{N,2} & \dots & 1 \end{vmatrix} \quad (10.20)$$

Where the σ_i are the volatilities of the underlying market variables¹³, the $\rho_{i,j}$ are the correlations between the market variables and the superscript T denotes the transpose of the matrix.

The VaR estimate for the portfolio is then simply given by

$$\text{VaR} = \text{NSD } \sigma_P \quad (10.21)$$

Example : Calculation of Delta VaR for a Portfolio of Two Options on Crude Oil and Natural Gas Futures

Consider a portfolio containing a one year at-the-money option on a two year oil future and a three month at-the-money option on a six month natural gas future on the 30 March 2000. We wish to calculate the Value-at-Risk for a one working day horizon with a confidence level of 95%. We assume interest rates are flat at 5%. Using our standard notation, with subscripts 1 and 2 denoting the oil and gas contracts respectively, we have the following:

$$K_1 = 21.21, T_1 = 1.00, s_1 = 2.00, F_1 = 21.21$$

$$K_2 = 2.99, T_2 = 0.25, s_2 = 0.50, F_2 = 2.99$$

$$T_h = 0.003968, p = 95\%, \text{NSD} = -1.65$$

We fit a three factor version of the multi-factor forward curve model described in chapter 8 to two years of historical NYMEX crude oil and natural gas futures data from 1 April

¹³ We note again that the volatilities must be computed to account for the variation of forward price return volatilities over the VaR horizon (see chapter 8).

1998 to 31 March 2000. Using this model the prices and deltas of the two options are:

$$C_1 = 0.6981, C_2 = 0.1246$$

$$\text{Delta}_1 = 0.4921, \text{Delta}_2 = 0.5146$$

The leverage weights are therefore calculated to be:

$$w_1 = 25.77923, w_2 = 3.630484$$

The standard deviations of the two underlying forwards at T_h and their correlation are:

$$\sigma_1 = 0.004461, \sigma_2 = 0.012015, \rho_{1,2} = 0.254295$$

We then evaluate the Σ vector and R matrix via equations (10.18) and (10.19);

$$\Sigma = |0.056589, 0.022449|$$

$$R = \begin{vmatrix} 1.0 & 0.254295 \\ 0.254295 & 1.0 \end{vmatrix}$$

Which gives the standard deviation and VaR of the portfolio as;

$$\sigma_P = 0.065972, \quad \text{VaR} = -1.65 \times 0.065608 = -0.108514$$

Figure 10.6 shows the computed portfolio return distribution, which is simply normally

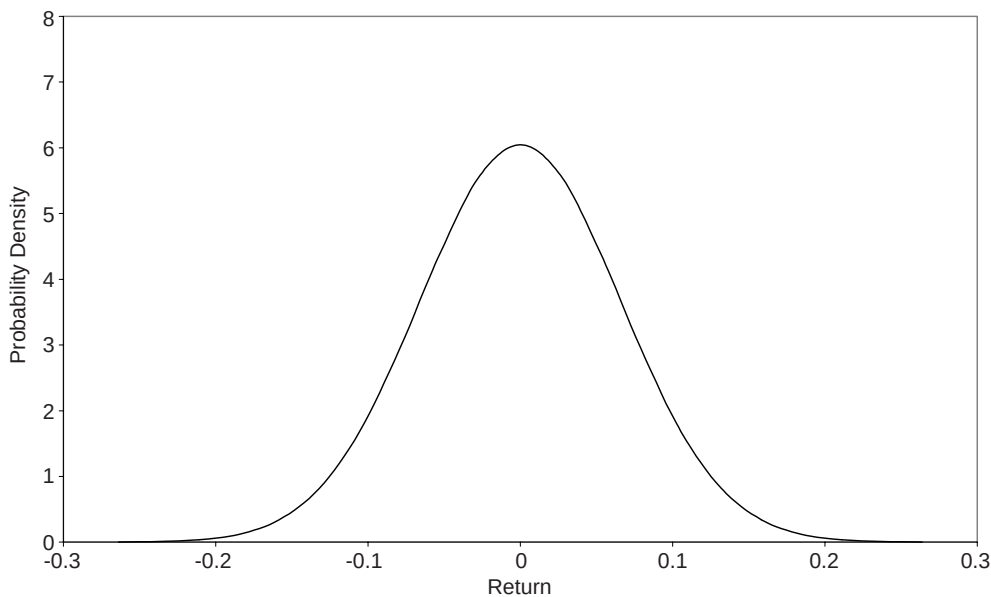


FIGURE 10.6 Delta VaR Example Portfolio Return Distribution

distributed with a standard deviation of 0.065608, for comparison with the methods discussed in the following sections.

10.6.2 Delta-Gamma VaR

The Delta-Gamma VaR method is an extension of the Delta VaR method described in the previous section which takes better account of the non-linear dependence of option prices on their underlying market variable. Figure 10.7 illustrates the difference between linear and non-linear instruments.

The value of a linear instrument as a function of the underlying variable can be represented by a straight line which is equivalent to a fixed amount of the underlying. For a non-linear instrument the relationship is a curved line and can't be represented as simply a fixed amount of the underlying variable.

The change in the value of an option (V) in terms of the changes in the underlying market variables can be represented by a Taylor series expansion;

$$\Delta V = \frac{\partial V}{\partial S} \Delta S + \frac{\partial V}{\partial t} \Delta t + \frac{1}{2} \frac{\partial^2 V}{\partial S^2} \Delta S^2 + \dots \quad (10.22)$$

In equation (10.22), $\frac{\partial V}{\partial t}$ is called theta and $\frac{\partial^2 V}{\partial S^2}$ is called gamma. When Delta VaR is used the option is converted into its delta equivalent position and only the first term

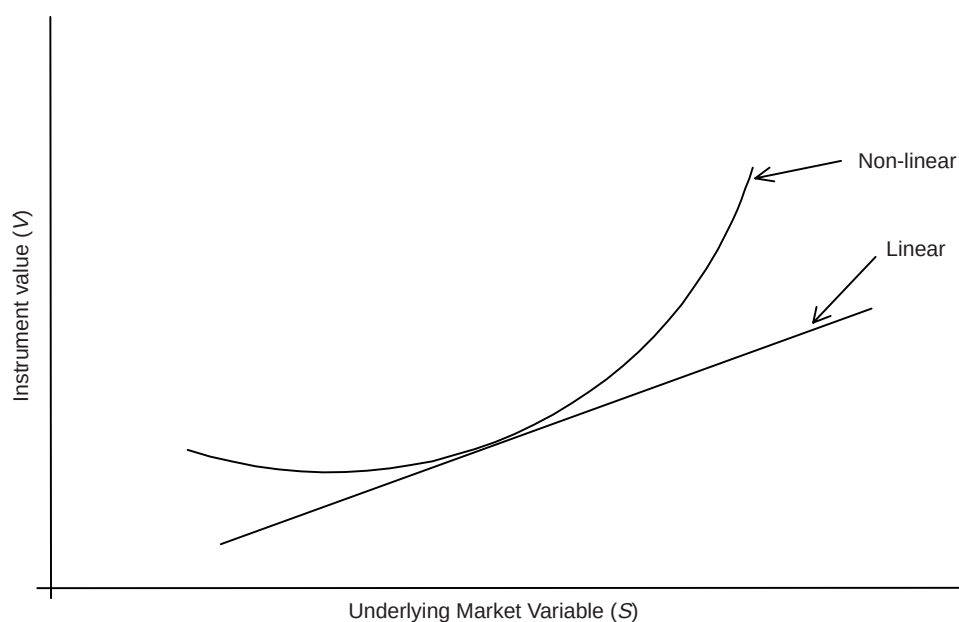


FIGURE 10.7 Illustration of the Difference Between Linear and Non-linear Instruments

in equation (10.21) is used giving the approximation for the change in the option value in equation (10.15). In Delta-Gamma VaR we retain all of the terms shown in equation (10.21)¹⁴ which gives a better approximation for the change in the instrument value as a function of the underlying asset;

$$\Delta V \approx \frac{\partial V}{\partial S} \Delta S + \frac{\partial V}{\partial t} \Delta t + \frac{1}{2} \frac{\partial^2 V}{\partial S^2} \Delta S^2 \quad (10.23)$$

Note that the theta term, $\frac{\partial V}{\partial t} \Delta t$, is also included but makes relatively little contribution to ΔV for typical VaR horizons. Equation (10.23) is often called the delta-gamma-theta approximation because of the inclusion of the theta term.

Representing the Portfolio Returns in Terms of the Underlying Returns

As for the Delta VaR method we have the following equation for the return on the portfolio (see equations (10.12) to (10.14));

$$\Delta P/P = (\Delta V_1 + \Delta V_2 + \dots \Delta V_N)/(V_1 + V_2 + \dots V_N) \quad (10.24)$$

Substituting for the change in the instrument values in equation (10.24) using equation (10.23) the return on the portfolio becomes¹⁵

$$\Delta P/P = \left(\frac{\partial V_1}{\partial S_1} w_1 x_1 + \frac{1}{2} \frac{\partial^2 V_1}{\partial S_1^2} w_1 S_1 x_1^2 + \frac{\partial V_1}{\partial t} w_1 T_h \right) + \dots \quad (10.25)$$

Note that equation (10.25) applies to the case of market instruments which are underlying market variables since in this case $\frac{\partial V}{\partial S} = 1$ and $\frac{\partial^2 V}{\partial S^2} = \frac{\partial V}{\partial t} = 0$

Computing Delta-Gamma VaR

Under the delta-gamma-theta approximation for the change in the instrument values the portfolio return is no longer a linear combination of underlying market variable returns as in equation (10.17). This in turn means that the portfolio return distribution is no longer normally distributed (which is true in reality and is the reason for using the more accurate delta-gamma-theta approximation). We cannot therefore compute the VaR estimate using the simple matrix multiplication we used for Delta VaR. A number of authors have suggested ways in which the delta-gamma-theta approximation can be used to actually compute a VaR estimate. The method proposed in the RiskMetrics Technical Document (4th Edition, December 1996, section 6.3) is to compute the first four moments of the portfolio distribution and then to fit a general distribution from the Johnson family (Johnson (1949)) using a version of an algorithm due to Hill, Hill and

¹⁴ Technically speaking we retain all terms up to second order.

¹⁵ The same approximation for returns, $x_i = \Delta S_i/S$, as in Delta VaR is used.

Holder (1976). No details of this approach are discussed in the RiskMetrics Technical Document and it would certainly be rather complex to implement with no guarantee of significantly increased accuracy. In an earlier RiskMetrics Monitor (First Quarter 1996) it is proposed that an approximation to the VaR estimate can be calculated in terms of the portfolio moments using the Cornish-Fisher expansion (see Johnson and Kotz (1970)). However, it is shown in the same article that this method has rather poor accuracy compared with the true VaR estimate computed by simulation. An excellent discussion and analysis of these and other methods can be found in Pritsker (1997).

The main point of the delta-gamma-theta approximation is that it allows us to compute the change in the value of an option without having to use the full option pricing model which may be relatively computationally intensive. In the full Monte Carlo simulation approach, which we discuss in the following section, the most computationally demanding part is the re-pricing of the options positions under each simulation. Our recommended Delta-Gamma approach is to use Monte Carlo simulation of the underlying market variables (which we discuss in section 10.6.3) together with the delta-gamma-theta approximation for the change in the option prices.

Example: Calculation of Delta-Gamma VaR for a Portfolio of Two Options on Crude Oil and Natural Gas Futures

TABLE 10.2 Example of Delta-Gamma Monte Carlo Simulation VaR Calculations

Sim.	F1	F2	C1	C2	Port	$\Delta P/P$
1	21.173	2.967	0.6791	0.1139	0.7931	-0.0361
2	21.325	2.953	0.7553	0.1069	0.8622	0.0480
3	21.056	2.938	0.6242	0.1002	0.7244	-0.1195
4	21.220	2.965	0.7024	0.1127	0.8151	-0.0093
5	21.147	2.958	0.6667	0.1092	0.7759	-0.0570
6	21.228	2.939	0.7062	0.1007	0.8069	-0.0193
7	21.260	2.971	0.7220	0.1159	0.8379	0.0184
8	21.147	2.931	0.6670	0.0971	0.7641	-0.0713
9	21.137	2.960	0.6618	0.1104	0.7723	-0.0614
10	21.261	2.957	0.7226	0.1089	0.8315	0.0107
...	...	...	...	...	...	...
491	21.147	3.065	0.6666	0.1680	0.8346	0.0144
492	21.174	2.949	0.6799	0.1053	0.7852	-0.0457
493	21.195	2.940	0.6901	0.1008	0.7910	-0.0386
494	21.281	2.988	0.7330	0.1245	0.8575	0.0422
495	21.253	2.957	0.7185	0.1089	0.8274	0.0057
496	21.223	2.970	0.7036	0.1153	0.8189	-0.0047
497	21.285	3.035	0.7350	0.1497	0.8847	0.0753
498	21.173	3.056	0.6793	0.1622	0.8416	0.0229
499	21.239	2.987	0.7118	0.1240	0.8358	0.0159
500	21.123	3.003	0.6555	0.1322	0.7877	-0.0426

We use the same portfolio, parameters and model as for the Delta VaR example in section 10.6.1.

The prices and sensitivities of the two options are:

$$\begin{aligned} C_1 &= 0.6981, C_2 = 0.1246 \\ \text{Delta}_1 &= 0.4921, \text{Delta}_2 = 0.5146 \\ \text{Gamma}_1 &= 0.2056, \text{Gamma}_2 = 1.2417 \\ \text{Theta}_1 &= -0.1979, \text{Theta}_2 = -0.1965 \end{aligned}$$

Table 10.2 shows an example of Delta-Gamma Monte Carlo simulation results. In Table 10.2, the columns headed “F1”, “F2”, “C1”, “C2”, “Port” and “ $\Delta P/P$ ” show the simulated futures prices, option prices computed using the delta-gamma-theta approximation, portfolio values, and portfolio returns respectively. The portfolio returns are defined as the difference between the simulated portfolio value and the current portfolio value divided by the current portfolio value.

The full simulated data set illustrated in Table 10.2 contains 500 daily portfolio returns - the VaR estimate for the chosen 95% confidence level is given by the $(1-0.95) \times 500$ sorted return i.e. the 25th sorted return which for this set of simulated data is -0.110426 . Figure 10.8 shows the resulting portfolio return histogram.

Notice that even for the relatively short one day horizon where Delta VaR should be quite accurate the VaR estimate under Delta-Gamma Monte Carlo VaR is 1.8% higher.

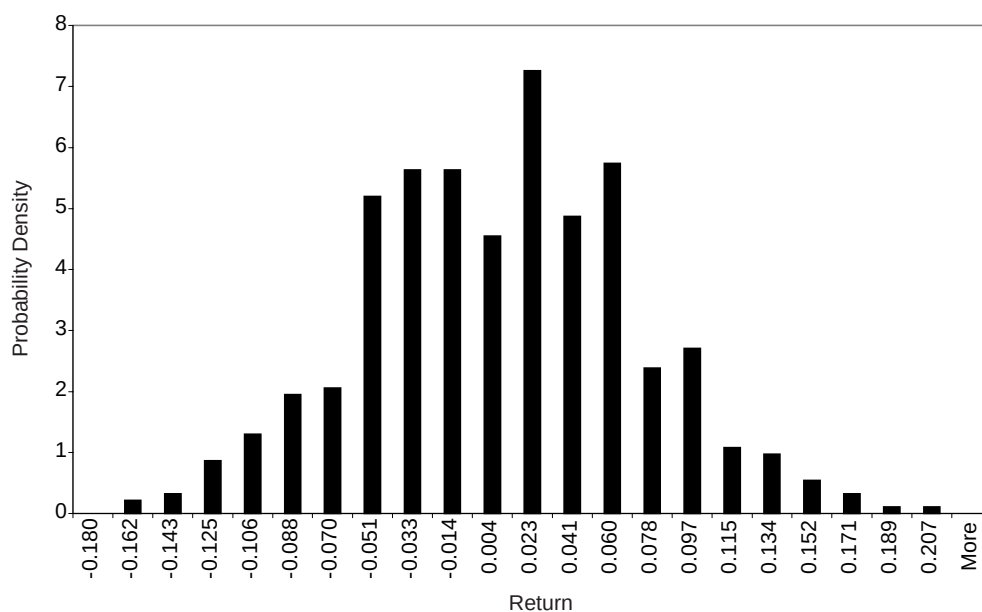


FIGURE 10.8 Delta-Gamma VaR Example Portfolio Return Distribution

We answer the question is this a more accurate estimate in the following section in which we compute VaR by full Monte Carlo simulation.

10.6.3 Monte Carlo Simulation

The Monte Carlo VaR method involves assuming a model for the joint behaviour of all relevant market variables and uses this to simulate future possible values of the portfolio. The model assumed is typically the same as for the Delta and Delta-Gamma VaR methods i.e. that returns are normally distributed. However, the advantages of this approach are that it allows for the accurate modelling of market behaviour such as the addition of jumps and incorporation of the knowledge of future events such as changes in the operation of a market. It also leads to a more accurate representation of the dependence of the prices of options on the market variables via the use of trading models. The problems with this approach are that it is relatively complex to understand and implement, and very computationally demanding.

The Monte Carlo approach requires the combined simulation of the spot and forward price curves underlying our portfolio at the VaR horizon. We can use the multi-factor model described in chapter 8 to model all the energy forward curves underlying the portfolio simultaneously. Recall from section 8.9.2 that the instantaneous returns in the different energy forward prices can be expressed in terms of a consistent set of volatility functions;

$$\frac{dF_e(t, T)}{F_e(t, T)} = \sum_{i=1}^n \sigma_{ei}(t, T) dz_i(t) \quad (10.26)$$

where e indexes the different energy forward curves e.g. oil, gas, electricity, etc. The volatility functions $\sigma_{ei}(t, T)$ describe how all the energy forward curves evolve together¹⁶. Under this model all forward price returns are jointly normally distributed and so in terms of the model for the underlying market variables it is identical to the Delta and Delta-Gamma VaR methods. However, it differs in that the dependence of non-linear derivatives on the market variables is more accurate. This is because option pricing models are used to compute the changes in the option prices for the each of the simulated states of the underlying forward curves. This is in contrast to the Delta and Delta-Gamma methods in which simple approximations are used to obtain the values of options as a function of the underlying forward curves.

In order to simulate the joint behaviour of the energy prices on which the portfolio depends we must obtain the volatility functions $\sigma_{ei}(t, T)$ from the covariance matrix of all the relevant forward prices. RiskMetrics supplies the following energy price volatility and correlation information:

¹⁶ An alternative approach would be to use the volatility functions for each energy forward curve. This would be equivalent to assuming the correlation between the different energy forward curves was zero.

	1m	3m	6m	1y
Natural Gas	✓	✓	✓	✓
Heating Oil	✓	✓	✓	✓
Unleaded Gas	✓	✓	✓	×
Light Sweet Crude	✓	✓	✓	✓

Therefore in order to evaluate long dated or other energy based instruments, further data must be obtained. This will typically be historical time series data which must be from reliable data sources and subjected to rigorous checks.

Using time series of daily returns the sample covariance matrix of the energy forward curves can be obtained from which the volatility functions can be obtained using Eigen Analysis (see section 8.6). The volatility functions allow us to compute the covariance matrix of the forward curves at the chosen VaR horizon (T_h):

$$\Sigma_{kl} = Cov[\ln(F_{e_k}(T_h, s_k)), \ln(F_{e_l}(T_h, s_l))] = \sum_{i=1}^n \left\{ \int_0^{T_h} \sigma_{e_{ki}}(u, s_k) \sigma_{e_{li}}(u, s_l) du \right\} \quad (10.27)$$

The VaR horizon may be longer than the original time series return period and therefore the original sample covariance matrix cannot be used. Furthermore, it is not correct to simply scale the original sample covariance matrix using the square root rule due to the time varying volatility of forward prices (i.e. mean reversion in the spot price) and also the fact that the volatility functions may not be Markovian (see chapter 8 for more details). We can then use Eigen Analysis (or for greater efficiency Cholesky Decomposition¹⁷) to obtain the “factors” required for simulating the returns at the horizon T_h . As we have seen in chapter 8 Eigen Analysis decomposes the covariance matrix as follows;

$$\Sigma = \Gamma \Lambda \Gamma^T$$

where

$$\Gamma = \begin{bmatrix} v_{11} & v_{21} & \dots & v_{m1} \\ v_{12} & v_{22} & \dots & v_{m2} \\ \dots & \dots & \dots & \dots \\ v_{1m} & v_{2m} & \dots & v_{mm} \end{bmatrix} \text{ and } \Lambda = \begin{bmatrix} \lambda_1 & 0 & 0 & 0 \\ 0 & \lambda_2 & 0 & 0 \\ 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & \lambda_m \end{bmatrix}$$

The columns of Γ are the eigenvectors and m is equal to the total number of forward prices to be simulated. We can then use these volatility functions to simulate the underlying forward prices at the VaR horizon many times and compute the changes in the value of the portfolio. The sample of portfolio changes give us a portfolio return distribution and thus the VaR estimate.

¹⁷ See RiskMetrics Technical Document, Appendix E

TABLE 10.3 Example of Monte Carlo Simulation VaR Calculations

Sim.	F1	F2	C1	C2	Port	DP/P
1	21.173	2.967	0.6791	0.1139	0.7931	-0.0361
2	21.325	2.953	0.7553	0.1069	0.8622	0.0480
3	21.056	2.938	0.6242	0.1003	0.7245	-0.1195
4	21.220	2.965	0.7024	0.1127	0.8151	-0.0093
5	21.147	2.958	0.6667	0.1092	0.7759	-0.0570
6	21.228	2.939	0.7062	0.1007	0.8069	-0.0192
7	21.260	2.971	0.7220	0.1159	0.8379	0.0184
8	21.147	2.931	0.6670	0.0971	0.7641	-0.0713
9	21.137	2.960	0.6618	0.1104	0.7723	-0.0614
10	21.261	2.957	0.7226	0.1089	0.8315	0.0107
...	...	...	...	...	...	...
491	21.147	3.065	0.6666	0.1680	0.8346	0.0144
492	21.174	2.949	0.6799	0.1053	0.7852	-0.0457
493	21.195	2.940	0.6901	0.1009	0.7910	-0.0386
494	21.281	2.988	0.7330	0.1245	0.8575	0.0422
495	21.253	2.957	0.7185	0.1089	0.8274	0.0057
496	21.223	2.970	0.7036	0.1153	0.8189	-0.0047
497	21.285	3.035	0.7350	0.1497	0.8847	0.0753
498	21.173	3.056	0.6793	0.1622	0.8415	0.0228
499	21.239	2.987	0.7118	0.1240	0.8358	0.0159
500	21.123	3.003	0.6555	0.1322	0.7877	-0.0426

Example: Calculation of Monte Carlo Simulation VaR for a Portfolio of Two Options on Crude Oil and Natural Gas Futures

The details of the portfolio are identical to the example for Delta VaR in section 10.6.1. The model for simulating the forward prices is the same three factor model used in the previous examples in sections 10.6.1 and 10.6.2. Table 10.3 shows a typical set of results.

In Table 10.3, the columns headed “F1”, “F2”, “C1”, “C2”, “Port” and “ΔP/P” contain the simulated futures prices, option prices, portfolio values, and portfolio returns respectively. The simulated option prices are obtained using the simulated futures prices and the three factor model estimated from historical data (see section 10.6.3). The portfolio returns are defined as in the previous section i.e. difference between the simulated portfolio value and the current portfolio value divided by the current portfolio value.

The full simulated data set illustrated in Table 10.3 contains 500 daily portfolio returns – the VaR estimate for the chosen 95% confidence level is given by the $(1-0.95) \times 500$ sorted return i.e. the 25th sorted return which for this set of simulated data is -0.110420. Figure 10.9 shows the resulting portfolio return histogram.

The VaR estimate under full Monte Carlo simulation is almost identical to that computed using the Delta-Gamma approach in the previous section. Full simulation can typically take between 100 and 1000 times longer to compute than Delta-Gamma Monte Carlo because of the full re-valuation of the option positions. Therefore, Delta-Gamma

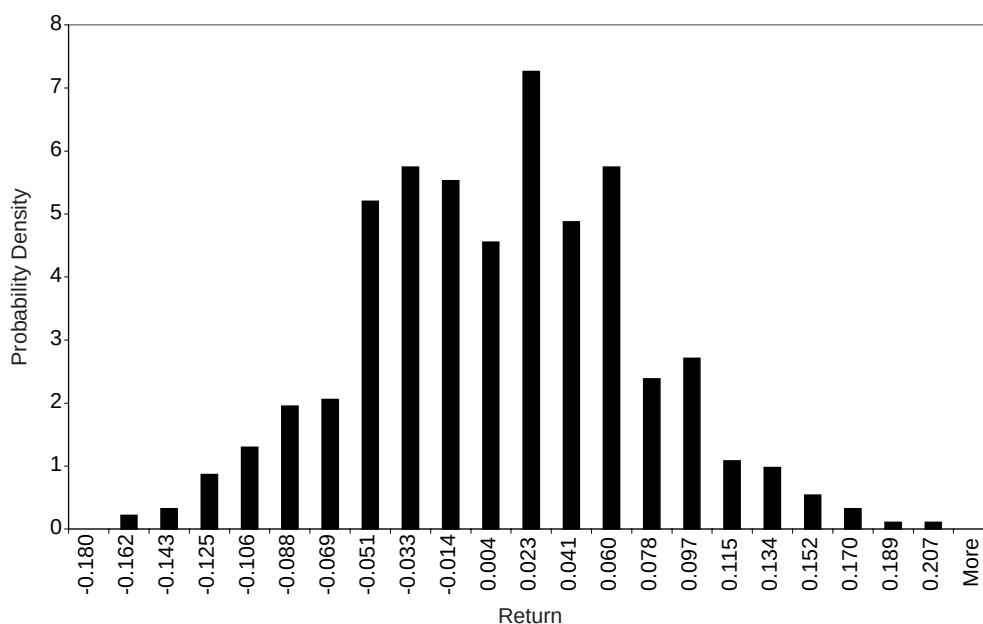


FIGURE 10.9 Monte Carlo Simulation VaR Example Portfolio Return Distribution

Monte Carlo is the recommended approach, particularly for shorter horizons. We saw in chapters 2, 3 and 7 that many energy prices exhibit jumps and stochastic volatility and that these factors can easily be incorporated into Monte Carlo simulations. Monte Carlo VaR therefore allows the incorporation of these important effects in a straightforward way.

10.6.4 Historical Simulation

The idea behind Historical Simulation VaR is to use historical time series' of market variables to obtain a distribution of portfolio returns. It therefore differs from Monte Carlo simulation which uses a model of the market variables and a set of random numbers to generate a portfolio distribution. For example, using the last two years of daily prices for the underlying market variables of our portfolio, we could compute the daily changes in the value of the current portfolio that would have occurred if the historical "changes" in the daily prices repeated themselves starting from today's values. This would give us approximately 500 daily returns for the current portfolio. Under the assumption that this past history of changes is a good forecast for the range of changes the portfolio could have in the future, we can compute a VaR estimate at our chosen confidence level.

The advantages of this approach are that it is simple to understand, intuitive and

relatively straightforward to implement. It can be applied to all instruments and all market risk types – all that is required is sufficient history of the relevant market variables. It allows the use of the pricing models used for trading to revalue option positions. The approach is particularly valuable if the user does not believe that the distributions of market returns can be described by normal distributions or tractable alternatives as is likely for energy markets.

The disadvantages of historical simulation are that it is computationally demanding due to the need to revalue the whole portfolio¹⁸. Furthermore, the portfolio distribution and thus the VaR estimate is completely determined by the distribution of the underlying market variables over the historical time period selected which can lead to unreasonable behaviour of the VaR estimate. For example, imagine we decide to use two years of daily changes and there is a large change in the underlying market variables two years ago. As we move forward in time and this particular day drops out of the historical data it is likely there will be a large change in the VaR estimate. A subtler problem is that the VaR estimate obtained will depend on the range of the market variables over the historical period used. If this period is too short then this may not reflect the range of possible values of the market in the future. Ideally we need a period of data which includes all possible changes in the market variables which could ever occur. Therefore, the period of data should be at least as long as the typical business cycle of the market.

Example: Calculation of Historical Simulation VaR for a Portfolio of Two Options on Crude Oil and Natural Gas Futures

We use the same portfolio as for the previous examples – the parameters values are given in the example in section 10.6.1. The prices of the two options¹⁹ are:

$$C_1 = 0.6981, C_2 = 0.1246$$

Table 10.4 shows a typical set of calculations. The first three columns in Table 10.2 contain the historical data – date and NYMEX closing futures prices on that date. The columns headed “F1” and “F2” contain the simulated futures prices and, since the VaR horizon is one working day, the simulated futures prices can be obtained by multiplying the current futures prices by the ratio of the historical futures price on consecutive days, as follows;

$$F_i(T_h) = F_i(0) \frac{\tilde{F}_i(t_{j+1})}{\tilde{F}_i(t_j)} \quad (10.28)$$

Where $\tilde{F}_i(t_j)$ is the historical futures price on date t_j . The columns headed “C1”, “C2”, “Port” and “ΔP/P” contain the simulated option prices, portfolio values, and portfolio

¹⁸ We can use the delta-gamma-theta approximation method to achieve more efficient evaluation in the same way as for Monte Carlo VaR.

¹⁹ We use the same three factor model as used in the example in section 10.6.1.

TABLE 10.4 Example of Historical Simulation VaR Calculations

Date	Oil(2y)	Gas(6m)	F1	F2	C1	C2	Port	DP/P
01/04/98	17.52	2.568	21.210	3.029	0.6974	0.1465	0.8438	0.0256
02/04/98	17.52	2.604	21.307	2.977	0.7460	0.1186	0.8646	0.0508
03/04/98	17.6	2.595	20.897	2.982	0.5533	0.1215	0.6748	-0.1798
06/04/98	17.34	2.591	21.247	3.106	0.7156	0.1935	0.9090	0.1049
07/04/98	17.37	2.694	21.381	3.006	0.7845	0.1337	0.9182	0.1161
08/04/98	17.51	2.711	21.077	2.953	0.6336	0.1070	0.7406	-0.0999
09/04/98	17.4	2.68	21.210	2.987	0.6974	0.1238	0.8212	-0.0019
10/04/98	17.4	2.68	21.039	2.837	0.6164	0.0606	0.6770	-0.1771
13/04/98	17.26	2.545	21.099	3.005	0.6442	0.1331	0.7773	-0.0553
14/04/98	17.17	2.56	21.482	3.010	0.8387	0.1362	0.9748	0.1848
...	...	...	...	...	...	...	...	...
20/03/00	20.92	2.84	21.291	2.933	0.7381	0.0980	0.8361	0.0162
21/03/00	21.13	2.864	21.423	3.012	0.8068	0.1372	0.9440	0.1474
22/03/00	20.82	2.893	20.899	3.017	0.5543	0.1400	0.6942	-0.1562
23/03/00	20.91	2.93	21.302	3.025	0.7433	0.1444	0.8877	0.0790
24/03/00	21.17	2.915	21.474	2.972	0.8343	0.1161	0.9504	0.1551
27/03/00	20.71	2.973	20.749	3.046	0.4926	0.1566	0.6492	-0.2110
28/03/00	20.35	3	20.841	3.014	0.5300	0.1382	0.6682	-0.1878
29/03/00	20.9	2.961	21.783	2.948	1.0127	0.1048	1.1175	0.3582
30/03/00	21.03	2.932						

returns respectively. The simulated option prices are obtained using the simulated futures prices and the three factor model estimated from the same historical data. The portfolio returns are defined as the difference between the simulated portfolio value and the current portfolio value divided by the current portfolio value.

The full data set in Table 10.2 contains 523 dates which gives 522 daily portfolio returns. The VaR estimate at the 95% confidence level is therefore the $(1-0.95) \times 522$ sorted return i.e. the 26th sorted return which is found to be -0.234053 . Figure 10.10 shows the resulting portfolio return histogram.

Comparing figures 10.6, 10.8, 10.9 and 10.10 it is clear that the shape of the historically simulated portfolio return distribution is somewhat different than the approximations obtained by Delta, Delta-Gamma and Monte Carlo VaR – there are two distinct reasons for this. Firstly, the return distribution of a portfolio containing call options is positively skewed due to the leverage effect of the options – small positive returns on the underlying lead to much larger positive returns for the portfolio. This can be seen in the right-hand tail of the histogram in Figure 10.10. Secondly, the historical futures price return distributions are fat-tailed leading to fat-tails in the portfolio return distribution. The left-hand tail in Figure 10.10 shows this quite clearly and it is also reflected in the VaR estimate (which depends on the shape of the left-hand tail of the distribution) being approximately twice as big as the other estimates.

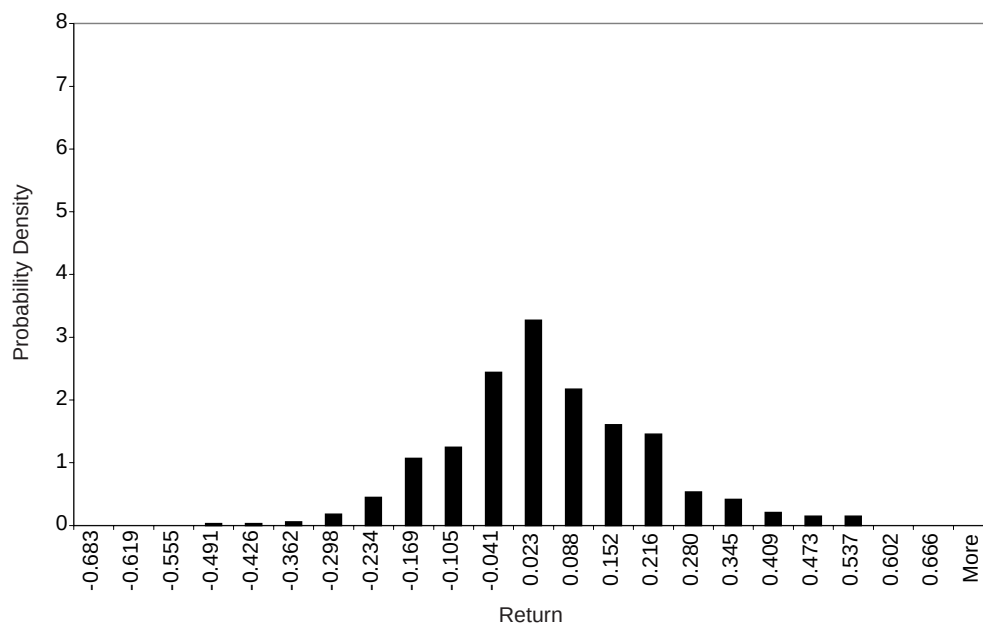


FIGURE 10.10 Historical Simulation VaR Example Portfolio Return Distribution

10.7 Testing Value-at-Risk Estimates

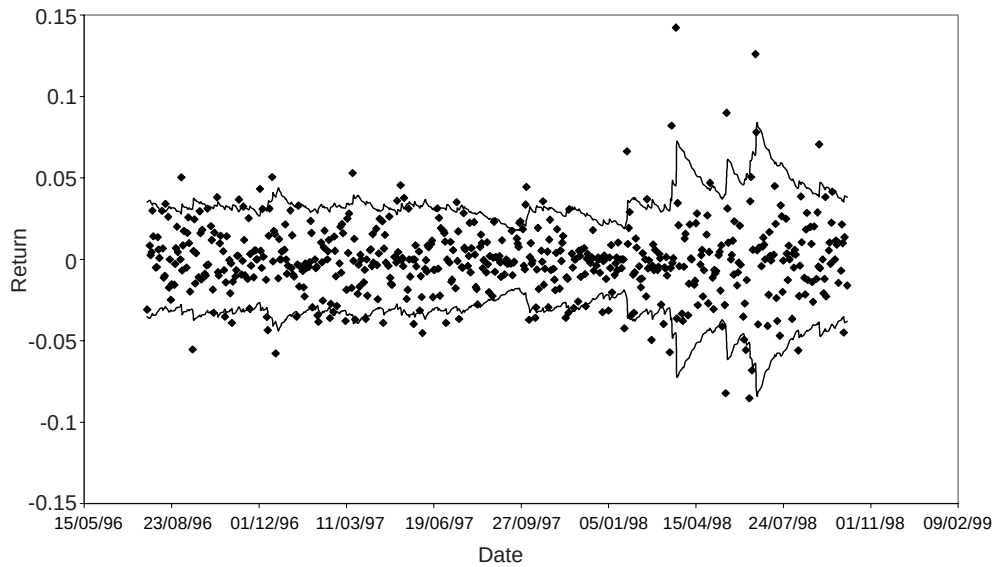
It is important that the VaR estimates are compared with the resulting market and portfolio returns to check how often actual returns exceed the VaR forecast – this process is called backtesting. The Bank for International Settlements (BIS) Basle Committee on Banking Supervision (Basle) has issued standards required for backtesting²⁰. Basle defines three zones depending on the number of exceptions in the previous year for a 99% confidence and a one day horizon. i.e. the number of times in the previous 250 working days the returns exceeded the expected number for the computed VaR level (i.e. for a period of 250 working days at a confidence level of 99% over one day, the expected number of returns exceeding the VaR level is 2.5 days).

Zone 1 or Green Zone: Four or less exceptions – the model is acceptable.

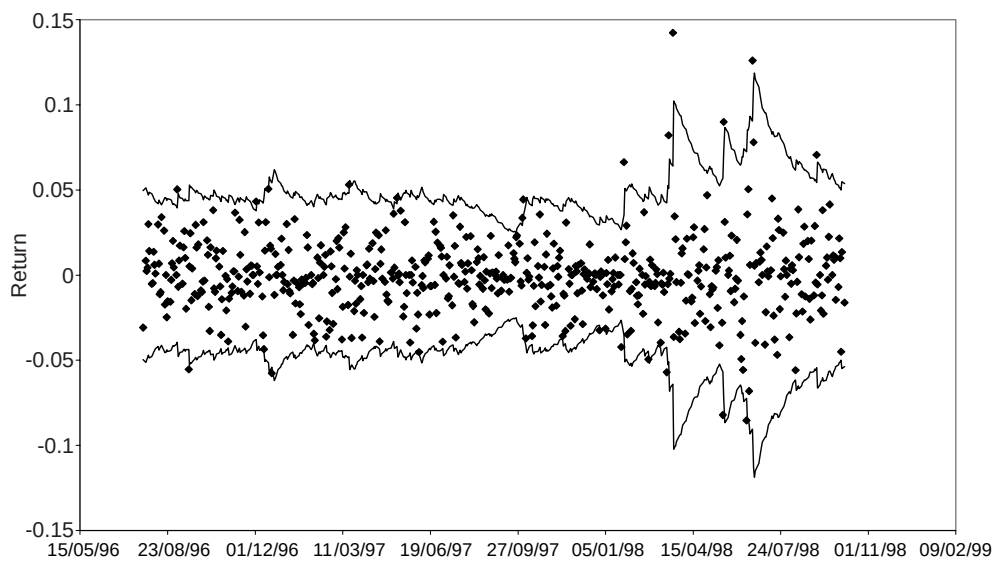
Zone 2 or Yellow Zone: Five to nine exceptions – certain actions are required such as increasing the safety multiplier.

Zone 3 or Red Zone: Ten or more exceptions – the model is unacceptable and must be substantially revised or changed.

²⁰ See the BIS website at www.bis.org and also the RiskMetrics Technical Document, Appendix F.

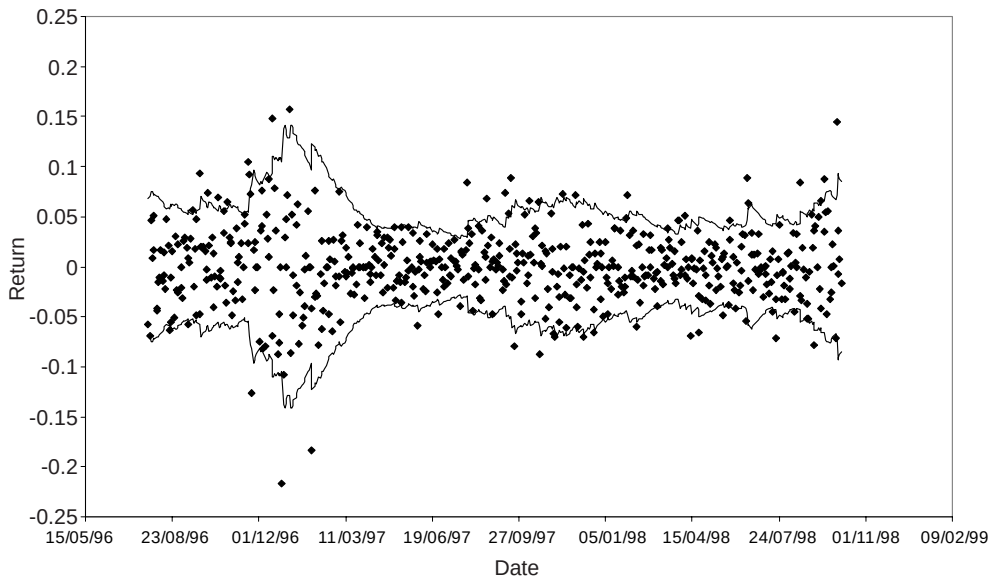


Key: Diamonds are daily returns, solid lines are 90% VaR Bands

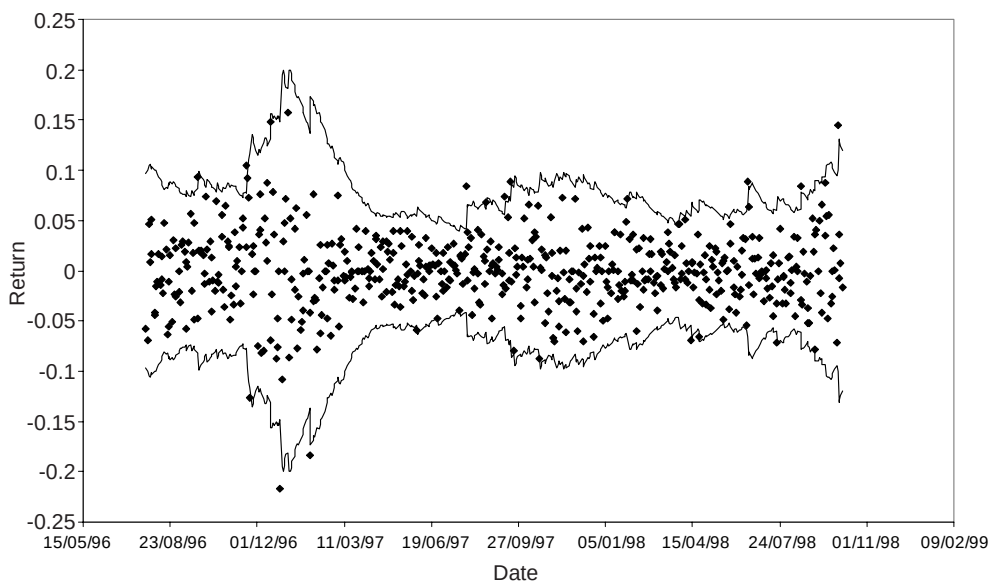


Key: Diamonds are daily returns, solid lines are 98% VaR Bands

FIGURE 10.11 Daily Returns, 90% and 98% VaR Bands for the Crude Oil Nearest Maturity NYMEX Futures Contract



Key: Diamonds are daily returns, solid lines are 90% VaR Bands



Key: Diamonds are daily returns, solid lines are 98% VaR Bands

FIGURE 10.12 Daily Returns, 90% and 98% VaR Bands for the Natural Gas Nearest Maturity NYMEX Futures Contract

In the following section we test the VaR forecasts for various underlying market variables.

10.7.1 Testing the VaR Forecast for the Market Variables

RiskMetrics VaR forecasts, for the market variable returns, are given by the bands of a confidence interval that is symmetric around zero. For example, the scale factors $-/+1.65$ give the 90% confidence interval or VaR level representing the 5th/95th percentiles of the standard normal distribution respectively. Under this assumption we expect 5% of the standardised observations²¹ to lie below -1.65 and 5% to lie above 1.65 . To test these VaR forecasts, for each day we compute the volatility forecast, $\hat{\sigma}$, using equation (10.6) with the RiskMetrics standard decay factor of 0.94 ²² and compare the VaR forecast at the 90% level i.e. $-/+1.65 \hat{\sigma}$ and at the 98% level i.e. $-/+2.33 \hat{\sigma}$. Figures 10.11 and 10.12 show the results, for the NYMEX crude oil and natural gas shortest maturity futures contracts respectively, over the period from July 1996 to September 1998.

In Figures 10.11 and 10.12 we expect that 5% of the returns will fall below $-1.65 \hat{\sigma}$ and 5% will fall above $1.65 \hat{\sigma}$. Similarly, we expect that 1% of the returns will fall below $-2.33 \hat{\sigma}$ and 1% will fall above $2.33 \hat{\sigma}$. For the crude oil Futures contract we find that 5.07% fall below $-1.65 \hat{\sigma}$, 6.64% fall above $1.65 \hat{\sigma}$, 2.10% fall below $-2.33 \hat{\sigma}$ and 1.57% fall above $2.33 \hat{\sigma}$. For the natural gas Futures contract we find 6.47% fall below $-1.65 \hat{\sigma}$, 5.07% fall above $1.65 \hat{\sigma}$, 2.10% fall below $-2.33 \hat{\sigma}$ and 1.57% fall above $2.33 \hat{\sigma}$. The conclusion is that at the 90% level the VaR estimates are quite good but at the more extreme 98% level they underestimate the VaR considerably.

10.8 Conclusions

In this chapter we discussed the main approaches to calculating VaR which are currently used and applied them to energy derivatives portfolios. The Delta and Delta-Gamma VaR methodologies rely for their relative simplicity on the assumption that the underlying market variables are normally distributed. As we have seen in the earlier chapters (see chapters 2, 3, 6, 8, and 9) this assumption may not be particularly good for energy markets – jumps in the spot price and random volatility lead to non-normal distributions for energy spot and forward price returns. Although the implementation of the Monte Carlo VaR approach, studied in this chapter, was also based on the assumption of normally distributed returns, we showed in chapters 2, 3 and 7 how Monte Carlo simulation can be extended to incorporate jumps and stochastic volatility. The evaluation of the VaR estimate in the Monte Carlo approach is not affected by adding these extra factors.

²¹ The standardised returns are the mean centred return normalised by its standard deviation.

²² We start computing the volatility forecasts after 90 days, as before this date we don't have enough data to compute a forecast.

In a recent study Marshall and Siegel (1997) tested the implementation and use in practice of one particular VaR model by assessing the variation in VaR estimates produced by different commercial implementations of the same model of value-at-risk. The authors asked a number of leading risk management system vendors to assess the value-at-risk of an identical portfolio of instruments of varying complexity according to the RiskMetrics methodology. The portfolios tested ranged from 'linear' instruments such as bonds, through interest rate swaps and FRA's to 'non-linear' Foreign exchange options and interest rate caps and floors.

The results show that for the "linear" instrument classes the VaR estimates were fairly consistent across vendors. However, for the more complex instruments the variations were very large – for example, for interest rate swaps the total VaR had a mean of US\$ 306,663 and a standard deviation of US\$ 66,648 (21%), for foreign exchange options – a mean total VaR value of US\$ 763,317 and standard deviation of US\$ 198,829 (25%), and interest rate caps had a mean VaR of US\$ 483,130 and standard deviation of US\$ 115,194 (28%).

It is therefore important that the relevant people in the institution are appropriately trained so they fully understand the implementation of VaR. Furthermore, the VaR system whether internally developed or purchased from a system vendor should be fully tested by independent consultants with proven expertise in VaR and derivatives risk management.

It is clear from our analysis and others that the simple VaR methods do not provide the accuracy required for energy markets. However, the more accurate methods such as Historical and Monte Carlo simulation are very time consuming and complex to implement. Our recommendation would be to develop the VaR implementation in stages, beginning with the simplest Delta VaR. From this it is possible to build up via Delta-Gamma VaR to one of the more sophisticated analytical or semi-analytical methods²³. An implementation of Historical Simulation VaR, which is relatively straightforward to implement, should always be available as a comparative check for the other methods.

In this chapter we have described in some detail the implementation of the key components of a VaR calculation for all of the main methodologies. This information together with the RiskMetrics Technical Document should provide the reader with the necessary tools to begin the process of building an appropriate VaR system for their needs. Implementing VaR is a difficult and time consuming exercise, it is, however, an essential part of the risk management process.

²³ For example, RiskMetrics Mixture of Normals method (see RiskMetrics Monitor Second Quarter 1996) or RiskMetrics Generalised Error Distribution method (see RiskMetrics Technical Document, Appendix B).